

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EC)

Nov 2012

EC 310 Optical Communication

Date: 07.11.2012, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Enlist main elements of an optical fiber communication link with necessary block diagram and also explain in brief. [04]
- (b) Consider a Multimode silica fiber that has a core refractive index $n_1=1.48$ and a cladding index $n_2=1.46$ calculate [03]
- (a) Critical angle
 - (b) Numeric Aperture NA
 - (c) Acceptance angle
- Q - 2 (a) Write down the difference between Step index fiber and graded index fiber. [03]
- (b) Define Bending loss and also explain types of bending loss occur in single mode fiber. [06]
- (c) A Double heterojunction InGaAsP LED emitting at a peak wavelength of 1310nm has radiative and non radiative recombination times of 30 and 100ns, respectively. The drive current is 40mA. Calculate [05]
- (a) Bulk recombination lifetime
 - (b) Internal quantum efficiency
 - (c) Optical power generated

OR

- Q - 2 (a) Draw neat figure and explain in brief. [06]
- (i) Surface emitting LED
 - (ii) Distributed feedback (DFB) Laser
- (b) Explain three basic optical confinement methods used for bounding laser lights in lateral direction. [04]
- (c) What are the possible lensing schemes used to improve optical source to fiber coupling efficiency? Discuss with necessary diagram. [04]

- Q - 3 (a) Define photon absorption, spontaneous and stimulated emission process occurs in Laser with neat sketch. [05]
- (b) Explain types of mechanical misalignment occur between two joined fibers in brief. [03]
- (c) Why the high impedance and trans-impedance amplifier are required? Explain in detail with detailed figure. [06]

OR

- Q - 3 (a) Define Fiber splicing. Enlist different splicing techniques and explain fusion splicing of optical fiber. [05]
- (b) Enlist principal requirement of a good connector design. [02]
- (c) Draw the block diagram for digital signal transmission for signal path through an optical data link. [07]

SECTION - II

- Q - 4 (a) Describe Mach-Zehnder interferometer with required sketch and equations. [05]
- (b) Define modal noise and mode partition noise. [02]
- Q - 5 (a) State the operating principles of pin photo-detector? Explain its working with necessary diagram and equations. [06]
- (b) For the wavelength range $1300\text{nm} < \lambda < 1600\text{nm}$, the quantum efficiency for InGaAs is around 90 percent find out responsivity for this range. [03]
- (c) Discuss the losses occurs at connectors and splices and in the fiber with respect to link power budget. [05]

OR

- Q - 5 (a) Write short note on Photo-detector Noise. [06]
- (b) A silicon avalanche photodiode has a quantum efficiency of 65 percent at a wavelength of 900nm. Suppose $0.5\mu\text{W}$ of optical power produces a multiplied photocurrent of $10\mu\text{A}$. Find the multiplication factor M. [03]
- (c) What are the losses occurred in 2x2 fiber coupler? Explain each of that with necessary figure. [05]
- Q - 6 (a) Draw the basic structure of SONET Frame and explain in detail. [07]
- (b) Write short note on OTDR. [07]

OR

- Q - 6 (a) Explain amplification mechanism with reference to EDFA. [07]
- (b) Sketch general configuration of an eye diagram with measurement parameters and explain in brief. [07]
